

```
madhumitha@localhost:~  
Downloads Music rahul.sh XYZ  
emp.txt opendir.c REC  
exec.c os_lab riya  
[madhumitha@localhost ~]$ vi bestfit.c  
[madhumitha@localhost ~]$ gcc bestfit.c  
[madhumitha@localhost ~]$ gcc bestfit.c -o bestfit  
[madhumitha@localhost ~]$ ./bestfir  
bash: ./bestfir: No such file or directory  
[madhumitha@localhost ~]$ ./bestfit  
Enter number of memory blocks:  
5  
Enter size of each block:  
100 200 300 400 500  
Enter number of processes:  
4  
Enter size of each process:  
123 124 125 126  
  
Process No.    Process Size    Block No.  
1              123            2  
2              124            3  
3              125            3  
4              126            4  
[madhumitha@localhost ~]$
```

```
madhumitha@localhost:~ — /usr/bin/vim bestfit.c
#include <stdio.h>

int main() {
    int m, n;

    // Input number of memory blocks
    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);

    int blockSize[m], remainingSize[m];

    // Input block sizes
    printf("Enter size of each block:\n");
    for (int i = 0; i < m; i++) {
        scanf("%d", &blockSize[i]);
        remainingSize[i] = blockSize[i];
    }

    // Input number of processes
    printf("Enter number of processes:\n");
    scanf("%d", &n);

    int processSize[n], allocation[n];
```

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Activities Terminal Apr 7 14:12

madhumitha@localhost:~ — /usr/bin/vim bestfit.c

```
#include <stdio.h>

int main() {
    int m, n;

    // Input number of memory blocks
    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);

    int blockSize[m], remainingSize[m];

    // Input block sizes
    printf("Enter size of each block:\n");
    for (int i = 0; i < m; i++) {
        scanf("%d", &blockSize[i]);
        remainingSize[i] = blockSize[i];
    }

    // Input number of processes
    printf("Enter number of processes:\n");
    scanf("%d", &n);

    int processSize[n], allocation[n];

    // Input process sizes
    printf("Enter size of each process:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &processSize[i]);
        allocation[i] = -1; // Not allocated initially
    }

    // First Fit Allocation
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            if (remainingSize[j] >= processSize[i]) {
                allocation[i] = j;
                remainingSize[j] -= processSize[i];
                break;
            }
        }
    }
}
```

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```
Activities Terminal Apr / 14:13
madhumitha@localhost:~ — /usr/bin/vim bestfit.c

for (int i = 0; i < m; i++) {
    scanf("%d", &blockSize[i]);
    remainingSize[i] = blockSize[i];
}

// Input number of processes
printf("Enter number of processes:\n");
scanf("%d", &n);

int processSize[n], allocation[n];

// Input process sizes
printf("Enter size of each process:\n");
for (int i = 0; i < n; i++) {
    scanf("%d", &processSize[i]);
    allocation[i] = -1; // Not allocated initially
}

// First Fit Allocation
for (int i = 0; i < n; i++) {
    for (int j = 0; j < m; j++) {
        if (remainingSize[j] >= processSize[i]) {
            allocation[i] = j;
            remainingSize[j] -= processSize[i];
            break;
        }
    }
}

// Output allocation
printf("\nProcess No.\tProcess Size\tBlock No.\n");
for (int i = 0; i < n; i++) {
    printf("%d\t%d\t", i + 1, processSize[i]);
    if (allocation[i] != -1)
        printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
}

return 0;
```

```
Activities Terminal Apr 7 14:40 madhumitha@localhost:~  
[madhumitha@localhost ~]$ ls  
ABC emp.txt opendir.c REC  
abc.sh exec.c os_lab riya  
ab.h fork.c os_lab1.c riyaa.sh  
algo kee os_lab.c roundrobin.c  
a.out mad Pictures sample.c  
'banks alg@'$'\n''mkdir banks' mad3 pid.c sample.txt  
bestfit madd pidoutput sjf.c  
bestfit.c madhu priority sujje  
Desktop madhu.sh Public Templates  
dir.c maith rahuk.sh Videos  
Documents mn rahul vms  
Downloads Music rahul.sh XYZ  
[madhumitha@localhost ~]$ vi besttfit.c  
[madhumitha@localhost ~]$ gcc besttfit.c -o besttfit  
[madhumitha@localhost ~]$ ./besttfit.c  
bash: ./besttfit.c: Permission denied  
[madhumitha@localhost ~]$ gcc besttfit.c -o besttfit  
[madhumitha@localhost ~]$ ./a.out  
Enter number of memory blocks:  
4  
Enter size of each block:  
12 13 14 15  
Enter number of processes:  
4  
Enter size of each process:  
23 24 25 26  
  
Process No. Process Size Block No.  
1 23 Not Allocated  
2 24 Not Allocated  
3 25 Not Allocated  
4 26 Not Allocated  
[madhumitha@localhost ~]$
```

```
#include <stdio.h>
```

```
int main() {
    int m, n;
    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);
```

```
    int blockSize[m], remaining[m];
```

```
    printf("Enter size of each block:\n");
```

```
    for (int i = 0; i < m; i++) {
        scanf("%d", &blockSize[i]);
        remaining[i] = blockSize[i];
    }
```

```
    printf("Enter number of processes:\n");
    scanf("%d", &n);
```

```
    int processSize[n], allocation[n];
```

```
    printf("Enter size of each process:\n");
```

```
    for (int i = 0; i < n; i++) {
        scanf("%d", &processSize[i]);
        allocation[i] = -1;
    }
```

```
    for (int i = 0; i < n; i++) {
        int bestIndex = -1;
```

```
        for (int j = 0; j < m; j++) {
            if (remaining[j] >= processSize[i]) {
                if (bestIndex == -1 || remaining[j] < remaining[bestIndex]) {
```

```

    allocation[i] = -1;
}

for (int i = 0; i < n; i++) {
    int bestIndex = -1;

    for (int j = 0; j < m; j++) {
        if (remaining[j] >= processSize[i]) {
            if (bestIndex == -1 || remaining[j] < remaining[bestIndex]) {
                bestIndex = j;
            }
        }
    }

    if (bestIndex != -1) {
        allocation[i] = bestIndex;
        remaining[bestIndex] -= processSize[i];
    }
}

printf("\nProcess No.\tProcess Size\tBlock No.\n");
for (int i = 0; i < n; i++) {
    printf("%d\t%d\t", i + 1, processSize[i]);
    if (allocation[i] != -1)
        printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
}

return 0;

```